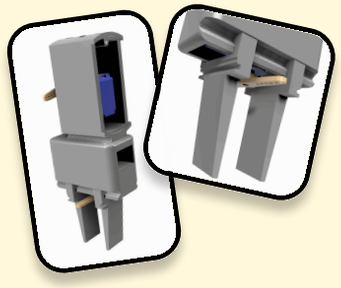
Autonomous Chess Player

by Akshay Khanna

Gripper



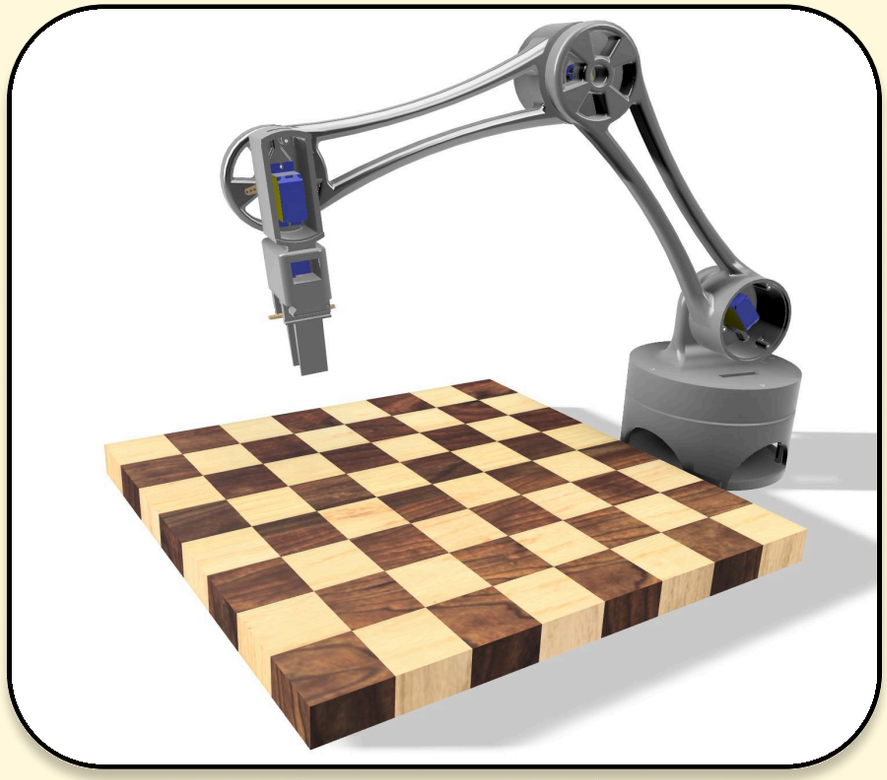
Actuated by a servo motor, the gripper hands are encircled by a rubber band which returns them to its original 'closed' position when the servo arm is at 90⁰.

Arms

Lightweight and modular design for the arms, to keep them compact and reusable. The lengths are calculated to ensure full reach over the board.

Base

Sturdy and robust base with design for assembly in mind. Housing a larger servo for greater balance and supporting the the weight of the full arm.



Evolution

Many iterations and 3D prints later...

Computer Vision

Chess board image recognition AI Model trained to detect white, black or empty squares.
Developed using Python and TensorFlow.

Chess Engine

Dedicated Python chess library integrated to suggest next move based on opponent moves.

Electronics

Simple open loop setup with servos and Arduino.

Roadmap

1. V1 Mech Design
2. V1 3D Printing
3. V1 Computer Vision
4. Electronics Wiring
5. V1 Manual Mode
6. V2 Autonomous Mode
7. V2.1 Auto Calibration Mode
8. V3 Custom Chess Engine
9. AI World Champion...?